Homework Assignment
Chapter 2: How Time and Interest Affect Money

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Instructions:

- For manual calculations, you must clearly write out the standard factor notation (e.g., $P = F(P/F, i, n)$) before computing the final numerical value.
- Using factors properly is key to solving engineering economy problems. Ensure you practice identifying the correct factor for each cashflow type.
- Assume all interest rates are compounded per period unless otherwise stated.

Exercise 1: Single Amounts and Uniform Series
A young engineer decides to save for a down payment on a house. She deposits \$5,000 today, and plans to deposit \$2,000 at the end of each year for the next 8 years. Her father supports her with a one time gift of \$5,000 in year 4. The account earns an interest rate of 6% per year.

1. Draw the cash flow diagram from the engineer's perspective.
2. Using standard factor notation, compute the Future Worth (F) of her savings at the end of year 8.
3. What is the total amount of interest she has earned over the 8 years?
4. **Optional Challenge:** Instead of a single gift in Year 4, her father decides to support her continuously with an arithmetic series. He gives her \$1,000 at the end of Year 1, and increases his gift by \$200 each subsequent year through Year 8. She immediately deposits these gifts into the account alongside her own deposits. Calculate her new final savings at the end of Year 8.

Exercise 2: Amortizing Loans and Capital Recovery
A small manufacturing company takes out a loan of \$150,000 to purchase new packaging machine. The loan is to be repaid in equal semi-annual installments over 5 years at an interest rate of 8% per year.

1. Draw the cash flow diagram from the **bank's** perspective.
2. Compute the required semi-annual payment (A) using the appropriate factor.
3. Create a running amortization schedule for the first 3 years. Determine the exact proportion of the 3rd payment that goes toward paying down the principal versus covering the interest.

Exercise 3: Composite Series, Arithmetic Gradient, and Excel
A supply chain company buys a new autonomous robot for the warehouse. The initial installation and setup cost is \$50,000 at Year 0. The system requires two types of maintenance:

- Routine mechanical maintenance costs \$1,500 at the end of Year 1, and this cost increases by \$300 each subsequent year.
- Software licensing and sensor calibration costs a flat \$2,500 every year starting at Year 1.

Additionally, at the end of Year 4, a major overhaul of the robotic arms is required, costing an extra \$8,000. The expected life of the line is 6 years, with no salvage value. The company's Minimum Attractive Rate of Return (MARR) is 7% per year.

1. Draw the combined cash flow diagram from the company's perspective.
2. Calculate the equivalent Present Worth (P) and Equivalent Uniform Annual Cost (AW) over the 6 years using standard factor notation.
3. Conceptually, should the equivalent Present Worth (P) and Annual Worth (AW) of this cash flow be considered inflows or outflows? Briefly explain why.

Exercise 4: Geometric Gradient Problem Structuring
NoVisionTech Inc., a rapidly growing tech company, relied increasingly on AI to automate tasks, eventually firing many of their human resources to cut costs. However, as their reliance on AI increased and their data volume scaled, the cloud computing costs skyrocketed. Eventually, the AI infrastructure cost became higher than having computer scientists doing the job. The CEO decided to rehire the team and support them with a regulated use of AI.

To budget for this new hybrid approach, the finance team projects that the optimized cloud hosting subscription will cost \$12,000 at the end of the upcoming year (Year 1). Moving forward, due to the vendor's adjustments and organic growth, this cost is expected to compound, increasing at a **geometric** rate of 6% each year through the end of a 10-year planning horizon. The company's investors expect a high rate of return, establishing the interest rate at 12% per year.

1. Draw the cash flow diagram for this scenario.
2. Calculate the Present Worth (P) of these subscription costs at Year 0.
3. If the company wanted to negotiate a flat annual contract (A) instead of escalating costs, what is the equivalent uniform annual amount they should agree to pay over the 10 years?

Exercise 5: Derivation of Factors
Understanding the derivation of engineering economy factors is crucial for correctly applying them to non-standard cash flows.

1. Derive the formula for the Uniform Series Compound Amount factor $(F/A, i, n)$. Begin by a clear cashflow diagram.

Exercise 6: Spreadsheet Basics (Single and Uniform Series)
You are comparing two different investment opportunities over a 15-year period.

- **Option A:** Requires an initial deposit of $20,000 today into an account paying 5% per year.
- **Option B:** Requires you to deposit $1,200 at the end of each month for 15 years into an account paying a nominal rate of 6% per year compounded monthly (i.e., 0.5% per month).

Using Excel's `FV` function:

1. Determine the future worth of Option A.
2. Determine the future worth of Option B.
3. Which option yields a higher future worth? Write down the exact Excel functions you used for both options.

Validate your results by going backwards using the `PV` function.

Exercise 7: Comprehensive Cash Flow Analysis
A civil engineering firm is evaluating a contract to build and maintain a small toll bridge. The financial projections are as follows:

- An initial construction cost of $2,500,000 at Year 0.
- Routine annual maintenance costs of $50,000 starting in Year 1, increasing by $5,000 each year.
- A major repaving cost of $300,000 occurs exactly at the end of Year 8.
- Toll revenues are projected to be $400,000 in Year 1, growing geometrically by 3% each year as traffic increases.
- The contract lasts for 15 years. At the end of Year 15, the bridge will be handed over to the municipality, but the firm will receive a final salvage payment of $500,000.

The firm's MARR is 10% per year.

1. Draw the net cash flow diagram for the 15-year project,.
2. Calculate the total equivalent Present Worth (P) of the project . Is the handover cost fair?
3. Calculate the Equivalent Uniform Annual Worth (AW) of the project.

Exercise 8: Present Worth of Contracts
A local newspaper headline blared, "Bo Smith Signs for $30 Million." The article revealed that Bo Smith, the former record-breaking running back, signed a $30 million package with his team. The terms of the contract were:

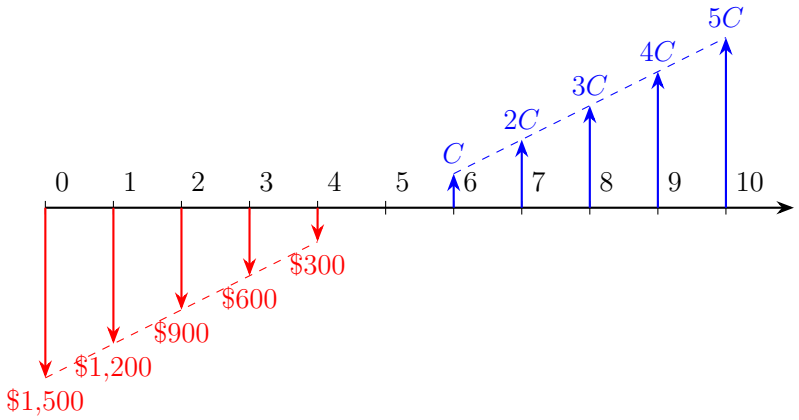
- $3 million paid immediately upon signing (Year 0).

- \$2.4 million per year for the first five years (with the first payment at the end of Year 1).
- \$3.0 million per year for the next five years (with the first payment at the end of Year 6).

If the interest rate is 8% compounded annually, what is the actual Present Worth of Bo's contract at the time of signing?

Exercise 9: Equipment Replacement and Viability
A company is considering replacing an old piece of industrial equipment to reduce operating and maintenance costs. The new equipment is quoted at \$215,000. After 12 years, the machine would have no salvage value, but it would save the company exactly \$35,000 in operating and maintenance costs at the end of each year. If the firm's Minimum Attractive Rate of Return (MARR) is 9% per year, is it economically justifiable to buy the new equipment? Calculate the equivalent Present Worth of the savings to justify your answer.

Exercise 10: Equating Complex Arithmetic Gradients
Consider the cash flow series given in the accompanying diagram.



What value of C makes the deposit series (upward arrows) equivalent to the withdrawal series (downward arrows) at an interest rate of 6% compounded annually?

MCQ
The following multiple-choice questions are designed to test your conceptual understanding and speed.

1. The (F/G) factor values can be derived by multiplying:
   (a) (P/F) and (A/G) factor values
   (b) (F/P) and (A/G) factor values
   (c) (P/F) and (P/G) factor values
   (d) (F/P) and (P/G) factor values
2. The value of the factor $(P/F, i, 10)$ can be found by getting the factor values for $(P/F, i, 4)$ and $(P/F, i, 6)$ and:

(a) Adding them

(b) Multiplying them

(c) Dividing them

(d) None of the above

3. At $i = 4\%$ per year, the equivalent annual worth A for years 1 through 6 of the following cash flows is closest to: (Year 1: \$800, Year 2: \$700, Year 3: \$600, Year 4: \$500, Year 5: \$400, Year 6: \$300)

(a) \$300

(b) \$560

(c) \$800

(d) \$1,040

4. An enthusiastic new engineering graduate plans to start a consulting firm by borrowing \$100,000 at 10% per year interest. The loan payment each year to pay off the loan in 7 years is closest to:

(a) \$18,745

(b) \$20,540

(c) \$22,960

(d) \$23,450

5. An engineer who believed in "save now and play later" wanted to retire in 20 years with \$1.5 million. At 10% per year interest, to reach the \$1.5 million goal, starting 1 year from now, the engineer must annually invest:

(a) \$26,190

(b) \$28,190

(c) \$49,350

(d) \$89,680

6. If you are able to save \$5,000 in year 1, \$5,150 in year 2, and amounts increasing by 3% each year through year 20, the amount you will have at the end of year 20 at 10% per year interest is closest to:

(a) \$60,810

(b) \$102,250

(c) \$351,500

(d) Over \$410,000